



Addressing Urban Infrastructure Issues: Solutions for Nairobi and Mexico City

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Traffic Congestion in Nairobi

Nairobi Problem: Matatu Congestion

What are Matatus?

Privately operated minibuses and buses

Main public transport in Nairobi

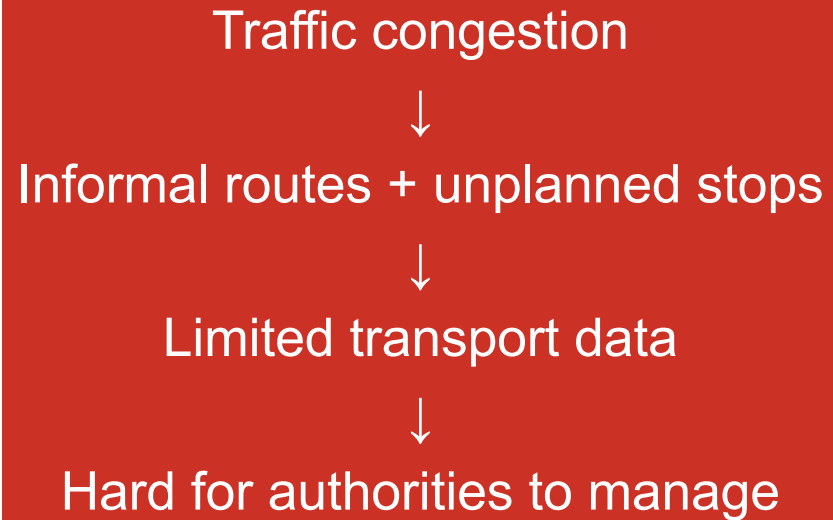


Image 1&2: Nairobi matatu&traffic congestion

Solutions

1. Passive Radio Frequency (RF) sensing hardware

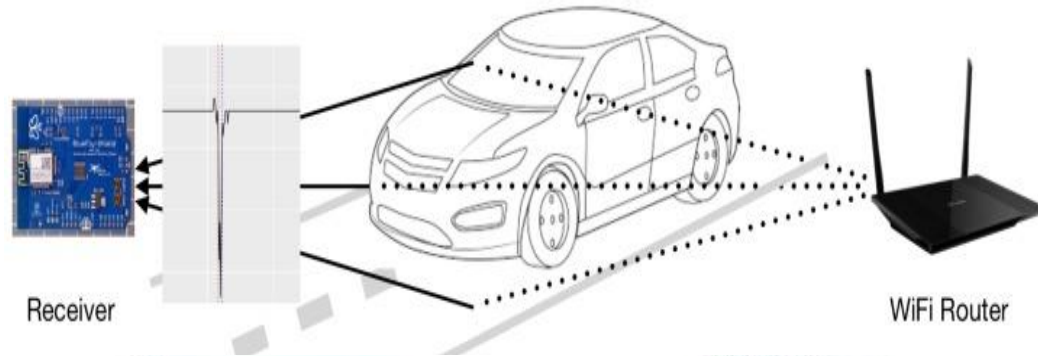


Image 3&4: Passive RF Sensing & deployment

2. app-based GPS tracking



Image 5: App-based GPS Tracking



Evaluation: Why Passive RF Sensing First?



Land Subsidence in Mexico City

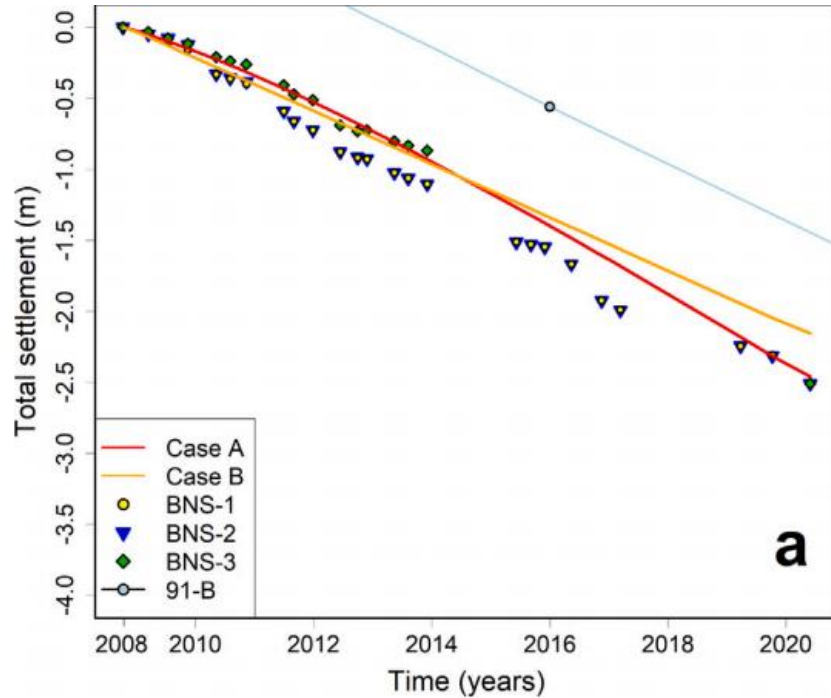


Image 6: Total settlement per year

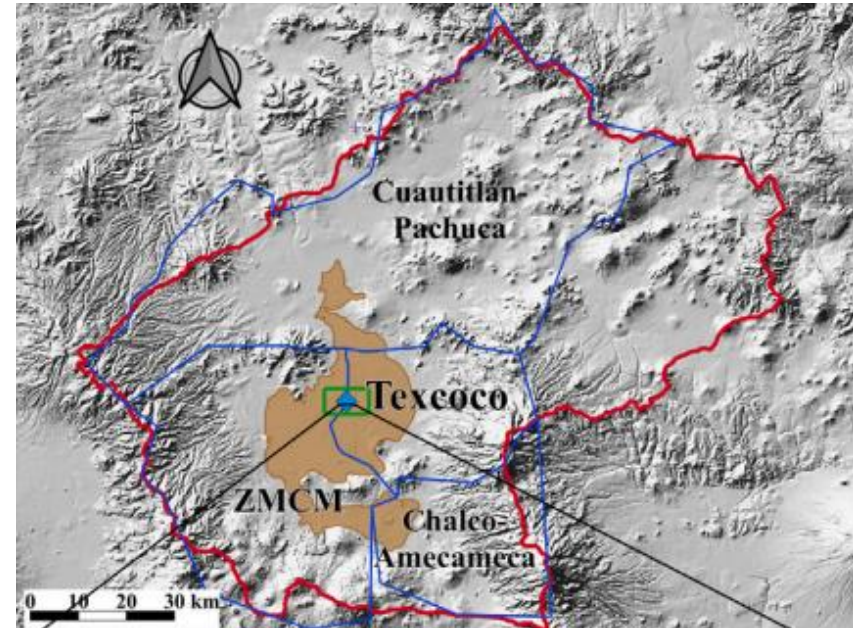


Image 7: Landform of Mexico City

- The average rate of subsidence reaches 0.314m/year
- The main reason for land subsidence is groundwater pumping

Solutions

1. Deep Soil Grouting

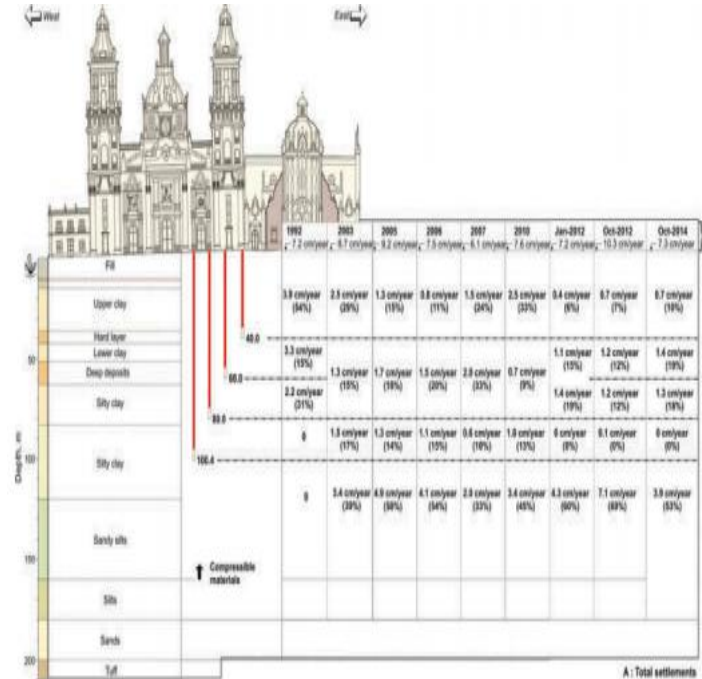
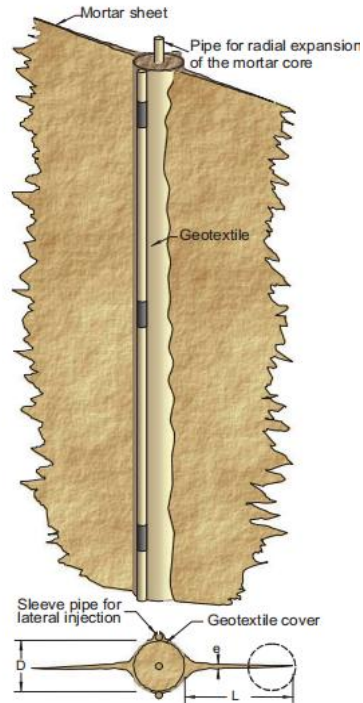


Image 8&9

2. Artificial Aquifer Recharge(AAR)

Wastewater collection and recycling



Artificial recharge of aquifer



Rising groundwater levels



Increasing the support force

Evaluations: Why AAR is better

- Scalable for city-wide deployment
- Lower long-term cost
- Tackle root hydrological cause
- Enhance future water security

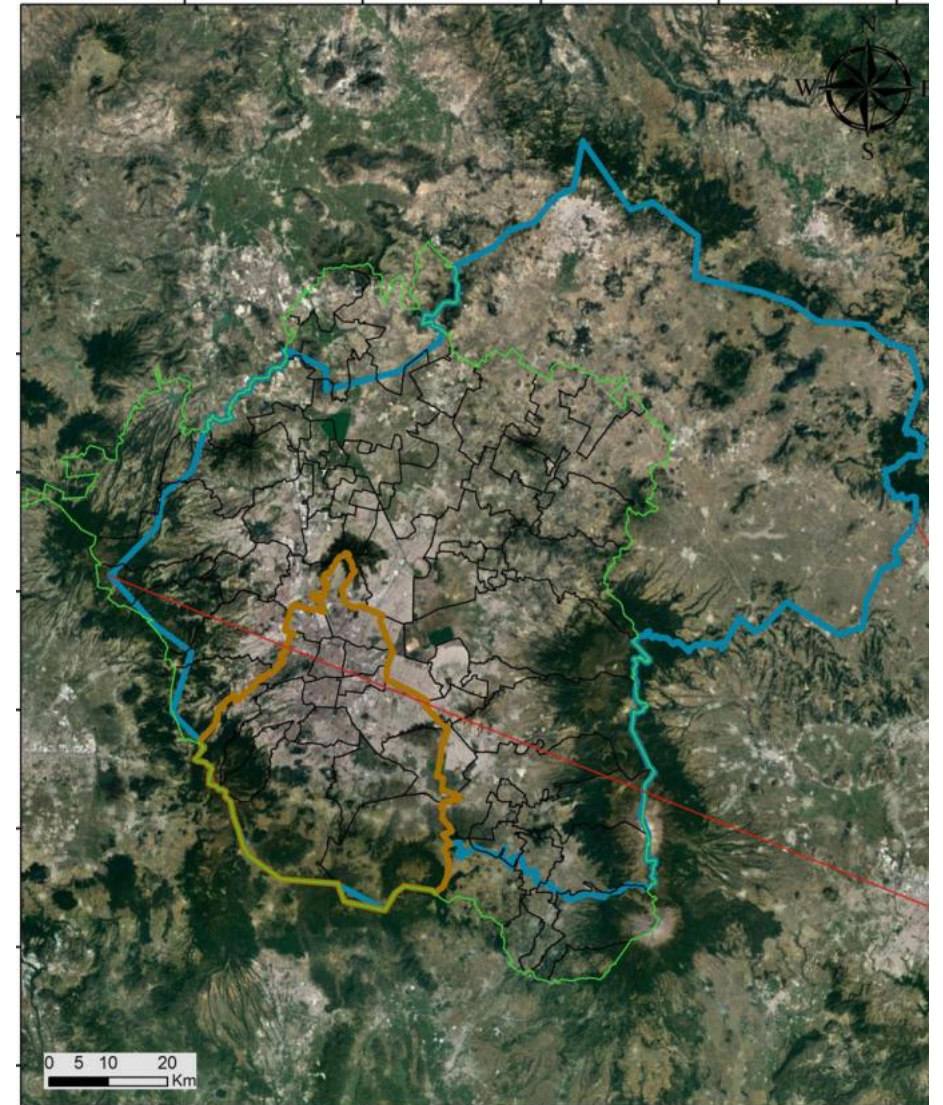


Image 10



Reflection: Different Problems, Similar Thinking

Visible problems

Congestion vs. subsidence

Hidden systems

Data gap vs. Groundwater loss

What I learned

Solution + wider city benefits

Future

A mature data platform for fairer
transport planning



Reflection: Engineering for EDI goals



- EDI insight: Digital Divide vs Resource Access
- Lesson: Technologies cannot ignore the vulnerable
- Future Goal: Solutions for everyone, not just the resourced



THANK YOU
for watching



Image Sources

[Image 1] CakeForBreakfast, “Matutu in motion in Nairobi, Kenya.jpg,” Wikimedia Commons. (2018). Accessed: Jun. 13, 2026. [Online]. Available:

https://commons.wikimedia.org/wiki/File:Matutu_in_motion_in_Nairobi,_Kenya.jpg. License: CC BY-SA 4.0.

[Image 2] MIT Civic Data Design Lab, “Congestion in Nairobi, Matatus make up the majority of vehicles on the roadway,” in “Digital Matatus,” Civic Data Design Lab. (2012–present). Accessed: Jun. 13, 2026. [Online].

Available: <https://civicdatadesignlab.mit.edu/Digital-Matatus>

[Image 3] S. Gupta, A. Hamzin, and A. Degbelo, “Illustration of deployment plan for the proposed hardware system,” in “A Low-Cost Open Hardware System for Collecting Traffic Data Using Wi-Fi Signal Strength,” Sensors, vol. 18, no. 11, Art. no. 3623, Oct. 2018, doi: 10.3390/s18113623. Accessed: Jun. 13, 2026. [Online].

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[Image 5] Wikimedia Commons, “Faraday emf experiment,” Wikimedia Commons. (2024). Accessed: Dec. 20, 2025. [Online]. Available:

https://upload.wikimedia.org/wikipedia/commons/thumb/2/2a/Faraday_emf_experiment.svg/640px-Faraday_emf_experiment.svg.png

[Image 4] S. Gupta, A. Hamzin, and A. Degbelo, “Experimental setup: Scenario 2 (Heavy traffic road: Steinfurter Straße),” in “A Low-Cost Open Hardware System for Collecting Traffic Data Using Wi-Fi Signal Strength,” Sensors, vol. 18, no. 11, Art. no. 3623, Oct. 2018, doi: 10.3390/s18113623. Accessed: Jun. 13, 2026. [Online].

Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6263683/>



Image Sources

[Image 5] MIT Civic Data Design Lab, “University of Nairobi students rode Matatus using an app to collect data,” in “Digital Matatus,” Civic Data Design Lab. (2012–present). Accessed: Jun. 13, 2026. [Online]. Available: <https://civicdatadesignlab.mit.edu/Digital-Matatus>

[Image 6&7] B. Zapata-Norberto, E. Morales-Casique, R. Contreras-Galvan, and J. A. Ramos-Leal, “Landform of Mexico City,” in “Land subsidence in Mexico City: New insights from field data and numerical modeling,” Groundwater for Sustainable Development, vol. 29, Art. no. 101432, Mar. 2025. Accessed: Jun. 13, 2026. [Online]. Available: <https://doi.org/10.1016/j.gsd.2025.101432>

[Image 8&9] E. Ovando-Shelley, E. Santoyo-Villa, and J. Hernández, figure from “Mexico City’s Metropolitan Cathedral and Sagrario Church 13 Years After Underexcavation and Soil Hardening,” International Journal of Architectural Heritage, vol. 10, nos. 2–3, pp. 346–359, 2016, doi: 10.1080/15583058.2015.1113331.

[Image 10] A. P. Nava, T. K. Parker, and R. B. C. Paredes, figure from “Challenges and experiences of managed aquifer recharge in the Mexico City Metropolitan Area,” Groundwater, vol. 60, no. 5, pp. 675–684, 2022, doi: 10.1111/gwat.13237.